Health-a-thon 2026

FINAL SUBMISSION

[Short subtitle explaining what your solution does]

BEAVERS

TEAM CODE [Insert team code]

Team Overview & Background

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Problem Statement and Supporting Evidence

01 Excessive Documentation & Manual Record Review

Clinicians spend significant time navigating records with different standard unit of measurement, reconstructing patient histories, and preparing medical documentation. According to a U.S. observational study, physician spend 49.2% [1] of their time in the EHR and desk work

[1] Sinsky et al. Physician time in ambulatory practice.

02 Clinician Burnout & Cognitive Overload

Clinicians face increasing mental and physical strain from repetitive documentation, prolonged EHR interactions, and constant switching between fragmented medical records. The need to manually reconstruct patient histories and identify critical information adds to cognitive fatigue,and can divert attention from direct patient care.

03 Consultation Amnesia & Medication Confusion

During consultations, expectant mothers struggle to retain verbal instructions, leading to critical advice going unheeded and confusion over medication dosages. According to a landmark clinical review, 40% to 80% [2] of medical information provided orally by doctors is forgotten immediately, with nearly half recalled incorrectly.

[2] Kessels. Patients' memory for medical information (2003).

04 The Inter-Visit Surveillance & Adherence Gap

While 96% of mothers receive paper MCP cards and 88% receive IFA tablets, physical records remain unindexed and inter-visit compliance drops drastically: only 44% take IFA for at least 100 days, and 41% fail to receive the WHO-recommended 4 ANC visits. Between monthly clinic visits, clinicians have zero visibility into daily physical activity, vitals, or emerging red flags like creeping blood pressure

[3] NFHS-5 (2019–21), Maternal Health.

05 Maternal Isolation & Perinatal Mental Health Gap

Expectant mothers lack safe, structured peer-support community spaces to share lived experiences, voice fears, and validate day-to-day pregnancy challenges. Perinatal depression affects 15% to over 30% of pregnant women in developing regions; without continuous inter-visit mood tracking and peer reassurance, acute distress goes unmonitored and stigmatized until reaching a clinical breaking point.

[4] WHO maternal mental health framework; perinatal peer-support trials.

Target Users and Use Case

Primary Users & Healthcare Settings



01. Expectant Mothers & Families

Rural, semi-urban and urban mothers across pregnancy and postpartum, with different literacy levels.

Core needs

Quick check-ins, clear medicine instructions and a safe peer community.

Pain relieved

Helps mothers revisit advice, follow care tasks and feel connected between visits.



02. Obstetricians & Antenatal Nurses

Public PHCs, district hospitals and private maternity clinics with busy antenatal services.

Core needs

A concise patient update and reviewable consultation documentation.

Pain relieved

Reduces repeated record review and manual drafting, supporting direct patient interaction.

Workflow Integration (How It Fits)

01 Pre-Visit Intake & Trend Synthesis

Mother logs check-ins, questions and health readings. OCR digitizes uploaded reports. An interval-scoped brief summarizes trends and patient questions.

02 Ambient Consultation & Clinician Review

Pair doctor and patient devices after recording consent. Transcribe the consultation, then review and edit the evidence-linked SOAP draft.

03 Post-Visit Care & Follow-Up

Share reviewed notes, prescriptions and follow-up tasks. Dose-linked daily medication reminders remain planned.

04 Between-Visit Monitoring & Peer Support

Review symptom and wearable trends. Screened peer discussions support connection, with configured professional review for concerning posts.

Solution Overview, Key Features & Impact

01 Continuous Prenatal Check-ins & Health Connect

- Weekly gestational-age reviews, quick comfort pulses and anytime concern logging.
- Capture symptoms, well-being, adherence and the mother's questions before each visit.
- Import steps, sleep, heart rate, BP and weight through permission-based Health Connect.
- Aggregate daily trends; red flags prompt guidance to contact the maternity care team.

02 Interval-Scoped AI Pre-Consultation Brief

- Scope the brief from the last completed visit to the current review.
- Gemini summarizes check-ins, concerns, vitals and wearable trends.
- Show priority flags, adherence and the mother's questions for clinician review.
- Rule-based fallback and on-demand refresh keep the brief inspectable.

03 Dual-Device Consultation & Evidence-Linked SOAP

- Doctor and patient devices record after consent; retry 15–25 s HTTPS chunks.
- Session + device + sequence keys prevent duplicate uploads; align both timelines.
- Filter crosstalk using overlap, Dice similarity and relative loudness.
- Sarvam / Whisper transcribe; Gemini / Qwen draft SOAP with transcript references.
- Clinicians edit, approve or reject; incomplete audio and uncertainty remain visible.

04 Lab OCR, Longitudinal Records & Prescription Support

- PaddleOCR digitizes reports; alias matching and unit conversion standardize metrics.
- Route uncertain values to verification; compare labs, vitals and maternity history.
- Gower similarity × coverage ranks reusable prescription templates.
- Inline completion populates editable medicine, dose, frequency and duration.
- Show known allergy and applicability flags; clinicians make prescribing decisions.

05 Explainable Hybrid Patient Similarity

- Algorithm: hybrid-gower-shrinkage-v2.
- Combine weighted Gower, severity / polarity-aware symptom matching and narrative cosine.
- Filter by gestational age; handle missing data and apply Bayesian coverage shrinkage.

$$G^* = [wG(G0 + 0.5c) + kS0] / (W + k)$$
$$S = (aG^* + bP^* + gN) / \text{active weights}$$

Weights: 45% structured, 35% symptoms, 20% narrative.
Show matches, conflicts, coverage and unchecked factors.
Descriptive retrieval aid; clinical validation is pending.

06 Natural-Language Workflow Automation

- English, Hindi and Hinglish requests navigate supported records and pages.
- Rules handle navigation; an Ollama / OpenAI-compatible LLM returns structured plans.
- An action registry, role checks and patient-context validation bound execution.
- Verify proposed field values and observed results; execute supported steps in order.
- Saving or sending requires confirmation; consent, recording and file selection stay manual.

07 Post-Visit Continuity & Screened Peer Support

- Revisit reviewed notes, prescriptions and follow-up tasks; complete checklists and log new concerns.
- Separate community identity, redact identifiers and route concerning posts to configured human review.

Expected impact

Less record assembly and drafting effort.
Earlier visibility of concerns and changing trends.
Clearer follow-up and continuity between visits.

End-to-End Workflow: Patient & Doctor Journeys

Patient Journey



01 Sign in & Set Up Profile

Accept the clinic invitation, sign in and complete the patient profile.

Screenshot
Sign In / Profile

02 Complete Check-ins

Log weekly reviews, quick symptom pulses, concerns, adherence and questions.

Screenshot
Check-in

03 Share Health Information

Upload lab reports and permit Health Connect imports where available.

Screenshot
Reports / Health Connect

04 Join & Consent to Consultation

Enter the room code, review consent and allow microphone access.

Screenshot
Join Consultation

05 View the Care Plan

Read the clinician's reviewed notes, prescriptions and follow-up tasks.

Screenshot
Care Plan / Visit History

06 Continue Care at Home

Complete tasks, log new concerns and join screened peer discussions.

Screenshot
Home / Community

Red-flag responses prompt guidance to contact the maternity care team promptly.

Doctor Journey



01 Open or Schedule the Patient Visit

Select the patient and appointment, or create a visit and share an invitation.

Screenshot
Patient List / Schedule

02 Review the AI Pre-Visit Brief

Review changes since the previous visit, priority flags and the mother's questions.

Screenshot
AI Brief

03 Verify Records & Comparable Cases

Check uncertain OCR, maternity history, similarity matches and missing factors.

Screenshot
Records / Similar Cases

04 Record the Consultation

Confirm consent, pair both devices and record. Stop capture to complete processing.

Screenshot
Recording

05 Review Notes & Prescription

Edit the cited SOAP draft and check medicine choices, allergy and applicability flags.

Screenshot
SOAP / Prescription

06 End the Visit & Share the Plan

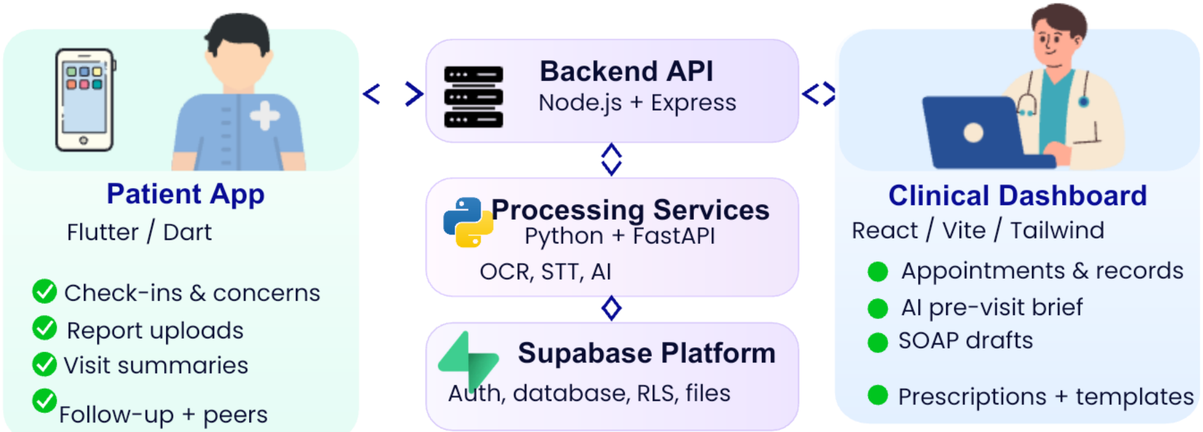
Confirm notes, prescriptions and tasks, then make the reviewed summary available.

Screenshot
Visit Summary

Patient inputs connect to clinician review and a shared, accessible care plan.

Technology, Tools & AI Approach

System Architecture & Key Technologies



AI Models, Algorithms & What They Enable

Lab report extraction	PaddleOCR, pdfplumber, PyMuPDF	Structured, standardized lab results
Consultation transcription	Sarvam Saaras v4 / Whisper (+ FFmpeg)	Aligned, de-duplicated transcripts
Pre-consultation summarization	Gemini (Flash) + rule-based fallback	Visit update, priority flags, questions
SOAP drafting & symptom extraction	Gemini / Qwen3-8B (optional)	Structured drafts with references
Patient similarity	Weighted Gower + symptom match + cosine + Bayesian shrinkage	Comparable cases with explained matches
Prescription template retrieval	Gower × coverage + template search	Reusable templates with allergy/conflict checks
Workflow automation	Rules-first + Ollama/OpenAI (Gemma 4 E4B)	Validated UI actions (open, fill, save)
Community screening	Local LLM + deterministic rules	Redaction, prioritization and human review

Integrations & Key Components

Android Health Connect
Steps, sleep, HR, BP, weight

HTTPS Audio Uploads
Retry by session, device and seq.

Socket.IO
Live status

Firebase Messaging
Community notifications

Email Invitations
Scheduled visits

In-app Follow-up
Check-ins + tasks

Redis
Audio cache

Review loop

1 Capture Reports, audio
2 Process OCR / STT / AI
3 Review Clinician
4 Share Care plan

Validation & Clinician Control

- **Access** — Role + RLS
- **Input** — Schema + limits
- **OCR** — Confidence, units
- **Audio** — Consent + sequence
- **AI Outputs** — Draft schema + citations
- **Similarity** — Coverage + shrinkage
- **Prescription** — Allergy + applicability
- **Automation** — Actions + confirmation
- **Community** — Screening + human review

Data Sources, Storage & Evaluation

Care Data
Check-ins, vitals, reports, maternity history, audio

Storage
Private buckets with controlled access

Dev & Evaluation
Synthetic fixtures, curated cases, seeded cohorts

Dataset Support
UCI maternal risk and MTS-Dialog (format support)

Build Sprint Journey and Progress

Four milestone phases evolved appointment preparation and report digitization into an integrated maternal-care prototype.

Phase 1: Connected Clinic Foundation

Appointments, access and shared records.

- Roles, appointments and invitations.
- Lab upload, OCR and verified results.
- Notes, prescriptions and visit history.

Core foundation integrated

Phase 2: Maternal Records & App

Structured maternity records and mobile access.

- Digital maternity and shared records.
- Investigations and vaccination history.
- Patient app and screened community.

Maternal-care workflows integrated

Phase 3: Consultation & AI

Capture, transcription and reviewable AI drafts.

- Dual-device Sarvam / Whisper STT.
- Cited SOAP and prescription templates.
- Gower, symptom and narrative matching.

Core consultation pipeline integrated

Phase 4: Continuity & Protection

Between-visit monitoring and controlled execution.

- Check-ins, pre-visit briefs and trends.
- Verified automation merged to master.
- Encrypted files and signed retrieval.

Integration achieved; live checks pending

Major Challenges and Changes

Challenge	Change made	Output delivered
Uncertain OCR values	Normalize units; verify uncertain results	Review before clinical reuse
Audio overlap and interrupted uploads	Align devices, retry chunks and expose gaps	More inspectable consultation capture
Unsupported AI details	Validate schema, citations and numeric claims	Editable, traceable clinical drafts
Automation reported unverified success	Check role, patient context and observed results	Controlled record and form actions
Sensitive documents and audio	Envelope encryption and expiring retrieval URLs	Protection for updated upload paths

Remaining Work Before Rollout

- Live-browser acceptance: merged assistant, confirmations, cancellation and multi-step actions.
- Physical-device testing: sustained audio, network recovery, wearables and notifications.
- Complete dose-linked reminders, batch cross-checks and speaker labelling.
- Validate key persistence, signed retrieval, recovery and historical-file handling.
- Run a clinician pilot to measure quality, usability, documentation effort and cost.

Now Functional

Appointments and shared records, lab OCR, digital maternity history and patient access.
Check-ins, wearable trends, dual-device transcription, AI briefs and reviewable SOAP drafts.
Patient similarity, prescription templates, screened community, automation and file encryption.

Milestone Outcome

A working integrated prototype with patient and clinician workflows.
Clinician review remains central across AI-assisted features.
Next milestone: real-device acceptance and a supervised clinical pilot.

Milestones describe repository integration. Clinical outcome validation remains pending.

Safety & Responsible AI

Core principle: Health-On flags, summarizes and supports review. Clinical decisions remain with the clinician.

01 Privacy & Meaningful Consent



- Record consultation consent, text version and acceptance time.
- Request microphone and Health Connect permissions explicitly.
- Separate community identity from the clinical identity.
- Use generic notifications and restrict clinical access.

Consent is recorded and permission-based; patient access stays controlled.

02 Security & Data Protection



- Authenticate users; enforce roles, ownership and row-level security.
- Use private document and audio storage with expiring retrieval URLs.
- Encrypt new files with AES-256-GCM; wrap file keys with RSA-OAEP.
- Keep audio logging focused on metadata rather than clinical content.

Deployment keys and retention need configuration. Historical plaintext files need separate assessment.

03 Visible, Reviewable AI Errors



- OCR, speech and language models can misread, omit or invent details.
- Route uncertain OCR to verification; expose audio gaps and failures.
- Check draft schema, transcript references and numeric claims.
- Use a rule-based brief fallback when model generation fails.

Checks reduce errors; they cannot guarantee factual accuracy or replace clinician review.

04 Human Oversight & Bounded Actions



- Store AI consultation outputs as drafts to edit, approve or reject.
- Keep prescription suggestions editable with allergy and applicability flags.
- Confirm supported save/send actions and verify their observed results.
- Keep recording, consent, destructive actions and file selection manual.

AI assists. Clinicians retain final decisions; concerning community content can reach configured human review.

05 Bias & Evidence Limits



- Similarity describes resemblance, with matches, conflicts and coverage.
- Distinguish missing information from an explicitly denied symptom.
- Template choices are preference examples, not evidence of efficacy.
- Test across gestational stages, languages, accents and patient groups.

Synthetic fixtures and format adapters do not establish clinical representativeness. Similarity remains unvalidated.

06 Accessibility & Inclusion



- Use guided check-ins, visual comfort ratings and short interaction paths.
- Support English, Hindi and Hinglish assistant requests.
- Retain manual reporting and navigation for users without wearables.
- Test screen readers, text scaling, literacy needs and regional-language comprehension.

Labeled controls and reduced motion help access. Full accessibility conformance is not yet established.

Clinical validation, subgroup evaluation, accessibility testing and deployment checks remain required.

Deployment, Cost & Sustainability Plan

Pilot: clinician web + shared API, Supabase and existing Android / PC hardware. Sarvam and Gemini stay usage-based.

Free infrastructure pilot
₹0 fixed hosting + API usage

Lean paid shared backend
₹3,380 / facility / month + APIs

Proposed lean API budget
≈ ₹10.52 / consultation

01 Free-Tier Pilot vs Lean Paid Setup

Cost component	Free-infrastructure pilot	Lean paid setup (proposed)
Backend + clinician web	Render Free + static frontend: ₹0/mo	Render Starter $7: ₹630/mo
Database, authentication, files	Supabase Free: ₹0/mo within limits	Supabase Pro $25: ₹2,250/mo
OCR + operations	Local PaddleOCR: ₹0 API; existing staff/PC	Local OCR + ₹500/mo operating allowance
Sarvam speech-to-text	20 billed audio min × ₹0.49 = ₹9.80/visit	Same: ₹9.80/visit
Gemini text processing	3.1 Flash-Lite, 40K in + 6K out: ₹1.71/visit	3.1 Flash-Lite, 20K in + 2K out: ₹0.72/visit
Shared fixed cost	**₹0/month**	**₹3,380/month**
Total variable API cost	**₹11.51/consultation**	**₹10.52/consultation**

02 Sarvam & Gemini API Rates

Sarvam pay-as-you-go

₹0.49 / billed audio minute.
₹100 signup credits are one-time; excluded from recurring budgets.

Gemini 3.1 Flash-Lite: planning model

$0.25 / 1M input + $1.50 / 1M output tokens.
40K input + 6K output = ₹1.71 per visit.

Gemini context optimization (proposed)

Bound total visit context to 20K input + 2K output.
At the same rates: ₹0.72 per visit.

Include thinking tokens and all calls; validate output quality.

03 Monthly Cost Sharing (220 Consultations per Doctor)

Doctors	Visits / month	Free-infra pilot facility total	Lean paid facility total	Lean paid per doctor
1	220	₹2,532	₹5,694	₹5,694
5	**1,100**	**₹12,661**	**₹14,952**	**₹2,990**
10	2,200	₹25,322	₹26,524	₹2,652

04 Keep Costs Low as Usage Grows

- Share one backend; avoid paid workspace seats unless deployment-team features are needed.
- Use existing OCR hardware, trim visit context and validate each optimized output.
- Meter audio/tokens and queue work; upgrade only when measured capacity requires it.

Setup: existing devices, ₹0 new hardware assumed. Staff/setup labour, electricity and GST excluded. Planning conversion: $1 = ₹90.
Free limits: Render sleeps on idle. Supabase: 500 MB DB / 1 GB files. Validate capacity, security and clinical readiness before a pilot.

Rates checked 7 Nov 2026. Figures are estimates; billed audio is counted across all devices. Validate security, capacity and output quality before a clinic pilot.

What Next? Plans to Take the Initiative Forward

Validate the workflow in clinical settings, extend interoperability and build a sustainable, responsible deployment.

01 Pilot, Validate & Build Partnerships



- Partner with an antenatal clinic or PHC for a supervised pilot.
- Start with a small clinician group and expand after feedback.
- Measure time, corrections, OCR/STT quality, participation and cost.
- Seek clinical collaborators, software partners and incubation support.

02 ABDM Integration & Verified Access



- Explore ABHA creation, verification and patient-record linking.
- Retrieve ABHA-linked records through patient-consented exchange.
- Explore HPR / HFR verification of professionals and facilities.
- Complete applicable sandbox, testing and security-review stages.

03 Expanded Roles & Team Workflows



- Add receptionist registration, appointment and queue workflows.
- Add nursing intake, vitals and consultation-preparation roles.
- Give facility administrators staff onboarding and access controls.
- Define permissions while preserving clinician authority.

04 Patient Similarity & Limited Predictions



- Expand verified longitudinal records and supported data formats.
- Improve missing-data handling, confidence and match explanations.
- Define narrow targets, such as follow-up needs, before prediction work.
- Require calibration, subgroup tests and independent validation.

05 Interoperability Adapter Toolkit



- Provide a documented API for hospital and laboratory software.
- Map identifiers, units, timestamps and record provenance.
- Validate imports, detect duplicates and preserve source details.
- Support relevant FHIR profiles as integrations mature.

06 Follow-Up, Language & Inclusion



- Add clinician-approved dose reminders and consented follow-up prompts.
- Improve task completion, interrupted uploads and weak-network recovery.
- Extend regional languages and test mixed-language consultations.
- Validate accessibility and usability across literacy levels.

Make the solution publicly available? Yes

Plan to share documentation, findings and known limitations on Health-a-thon and an open project site after the pilot.
An ABHA number alone does not grant record access; patient consent remains necessary.